notebook

January 3, 2024

[ ]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind
from joypy import joyplot
from scipy.stats import ks_2samp
from sklearn.cluster import KMeans

sns.set_palette("Spectral")
```

[ ]:
```
goalies = pd.read_csv('data\\nhlkaggle\game_goalie_stats.csv')
skater_stats = pd.read_csv('data\\nhlkaggle\game_skater_stats.csv')
teams_stats = pd.read_csv('data\\nhlkaggle\game_teams_stats.csv')
games = pd.read_csv('data\\nhlkaggle\game.csv')
teams = pd.read_csv('data\\nhlkaggle\\team_info.csv')
winners = pd.read_csv('data\\nhlkaggle\playoff_winners.csv')
```

0.0.1 Collection of processing code

[ ]:
```
#create year column for games
games['date_time_GMT'] = pd.to_datetime(games['date_time_GMT'])
games['year'] = games['date_time_GMT'].dt.year

teams_stats['year'] = teams_stats['game_id'].astype(str).str[:4]



#remove unknown team IDs
unknown = [88, 87, 90, 89]
teams_stats = teams_stats[~teams_stats['team_id'].isin(unknown)]
goalies = goalies[~goalies['team_id'].isin(unknown)]
skater_stats = skater_stats[~skater_stats['team_id'].isin(unknown)]

#apply team names for readability, team name to ID dict
teamdict = dict(zip(teams.team_id, teams.teamName))
reverseteamdict = dict(map(reversed, teamdict.items()))

#various dictionaries
seasondict = pd.Series(games['season'].values, index=games['game_id']).to_dict()
```

```
teams_stats['season'] = teams_stats['game_id'].map(seasondict)
skater_stats['season'] = skater_stats['game_id'].map(seasondict)


#list of all team names for iteration
teamnamelist = teams.teamName.to_list()
teams_stats['team_name'] = teams_stats['team_id'].replace(teamdict)
winners['team_id'] = winners['team_name'].map(reverseteamdict)
winners['team_id'] = winners['team_id'].fillna(0)
winners['team_id'] = winners['team_id'].astype('int64')
windict = pd.Series(winners['team_id'].values, index=winners['season']).
 ↪to_dict()


#overall goal distribution
allgoals = teams_stats.groupby(['team_name'])['goals'].sum()
allgoals.to_csv('allgoals.csv')

#overall avg goal distribution
avggoals = teams_stats.groupby(['team_name'])['goals'].mean()
avggoals.to_csv('avggoals.csv')

#remove coaches who only coached a few games
removals = teams_stats['head_coach'].value_counts().reset_index()
removals = removals[removals['head_coach']>10]['index'].values
teams_stats = teams_stats[teams_stats['head_coach'].isin(removals)]

#drop games of type 'A'
dropgames = games[games['type'] == 'A'].index
games.drop(dropgames, inplace=True)

#classify teams who won playoffs and players who were on winning teams
teams_stats['winningteam'] = teams_stats['season'].map(windict)
teams_stats['winseason'] = (teams_stats['team_id'] ==␣
 ↪teams_stats['winningteam']).astype(int)
teams_stats.drop('winningteam', axis=1, inplace=True)

skater_stats['winningteam'] = skater_stats['season'].map(windict)
skater_stats['winseason'] = (skater_stats['team_id'] ==␣
 ↪skater_stats['winningteam']).astype(int)
skater_stats.drop('winningteam', axis=1, inplace=True)

#change some data to numeric
teams_stats['hometeam'] = teams_stats['HoA'] == 'home'
teams_stats[['hometeam', 'won']] = teams_stats[['hometeam', 'won']].astype(int)
```

```
#drop unneeded columns
games_cols = ['game_id', 'season', 'type', 'date_time_GMT', 'away_team_id',␣
↪'home_team_id', 'away_goals', 'home_goals', 'outcome', 'venue', 'year']
games = games[games_cols]

teams_cols = ['game_id', 'team_id', 'hometeam', 'won', 'settled_in',␣
↪'head_coach', 'goals', 'shots', 'team_name', 'season', 'winseason']
teams_stats = teams_stats[teams_cols]
# dummies = pd.get_dummies(teams_stats['head_coach'])
# teams_stats = pd.concat([teams_stats, dummies], axis=1)
# teams_stats = teams_stats.drop(['head_coach'], axis=1)

skater_stats = skater_stats[['game_id', 'player_id', 'team_id', 'timeOnIce',␣
↪'assists', 'goals', 'shots', 'season', 'winseason']]

teams = teams.drop('link', axis=1)
```

[ ]:
```
games.head()
```

[ ]:
```
      game_id    season type              date_time_GMT  away_team_id  \
0  2016020045  20162017    R 2016-10-19 00:30:00+00:00              4
1  2017020812  20172018    R 2018-02-07 00:00:00+00:00             24
2  2015020314  20152016    R 2015-11-24 01:00:00+00:00             21
3  2015020849  20152016    R 2016-02-17 00:00:00+00:00             52
4  2017020586  20172018    R 2017-12-30 03:00:00+00:00             20

   home_team_id  away_goals  home_goals       outcome           venue  year
0            16           4           7  home win REG  United Center  2016
1             7           4           3   away win OT  KeyBank Center  2018
2            52           4           1  away win REG     MTS Centre  2015
3            12           1           2  home win REG      PNC Arena  2016
4            24           1           2  home win REG    Honda Center  2017
```

[ ]:
```
winners
```

[ ]:
```
      season   team_name        head_coach  team_id
0   20002001      Devils    Larry Robinson        1
1   20012002   Avalanche       Bob Hartley       21
2   20022003   Red Wings     Scotty Bowman       17
3   20032004      Devils         Pat Burns        1
4   20042005   Lightning   John Tortorella       14
5   20052006        None              None        0
6   20062007  Hurricanes  Peter Laviolette       12
7   20072008       Ducks     Randy Carlyle       24
8   20082009   Red Wings      Mike Babcock       17
9   20092010    Penguins        Dan Bylsma        5
10  20102011  Blackhawks  Joel Quenneville       16
```

```
11  20112012       Bruins     Claude Julien        6
12  20122013        Kings      Daryl Sutter       26
13  20132014  Blackhawks  Joel Quenneville       16
14  20142015        Kings      Daryl Sutter       26
15  20152016  Blackhawks  Joel Quenneville       16
16  20162017     Penguins     Mike Sullivan        5
17  20172018     Penguins     Mike Sullivan        5
18  20182019     Capitals       Barry Trotz       15
19  20192020        Blues      Craig Berube       19
```

[ ]:
```
teams_stats.head()
```

[ ]:
```
      game_id  team_id  hometeam  won settled_in        head_coach  goals  \
0  2016020045        4         0    0        REG      Dave Hakstol    4.0
1  2016020045       16         1    1        REG  Joel Quenneville    7.0
2  2017020812       24         0    1         OT     Randy Carlyle    4.0
3  2017020812        7         1    0         OT      Phil Housley    3.0
4  2015020314       21         0    1        REG       Patrick Roy    4.0

   shots   team_name    season  winseason
0   27.0      Flyers  20162017          0
1   28.0  Blackhawks  20162017          0
2   34.0       Ducks  20172018          0
3   33.0      Sabres  20172018          0
4   29.0   Avalanche  20152016          0
```

[ ]:
```
skater_stats.head()
```

[ ]:
```
      game_id  player_id  team_id  timeOnIce  assists  goals  shots    season  \
0  2016020045    8468513        4        955        1      0      0  20162017
1  2016020045    8476906        4       1396        1      0      4  20162017
2  2016020045    8474668        4        915        0      0      1  20162017
3  2016020045    8473512        4       1367        3      0      0  20162017
4  2016020045    8471762        4        676        0      0      3  20162017

   winseason
0          0
1          0
2          0
3          0
4          0
```

[ ]:
```
teams.head()
```

[ ]:
```
   team_id  franchiseId     shortName    teamName abbreviation
0        1           23    New Jersey      Devils          NJD
1        4           16  Philadelphia      Flyers          PHI
```

```
2        26            14   Los Angeles      Kings           LAK
3        14            31     Tampa Bay  Lightning           TBL
4         6             6        Boston     Bruins           BOS
```

[ ]:
```
goalies.head()
```

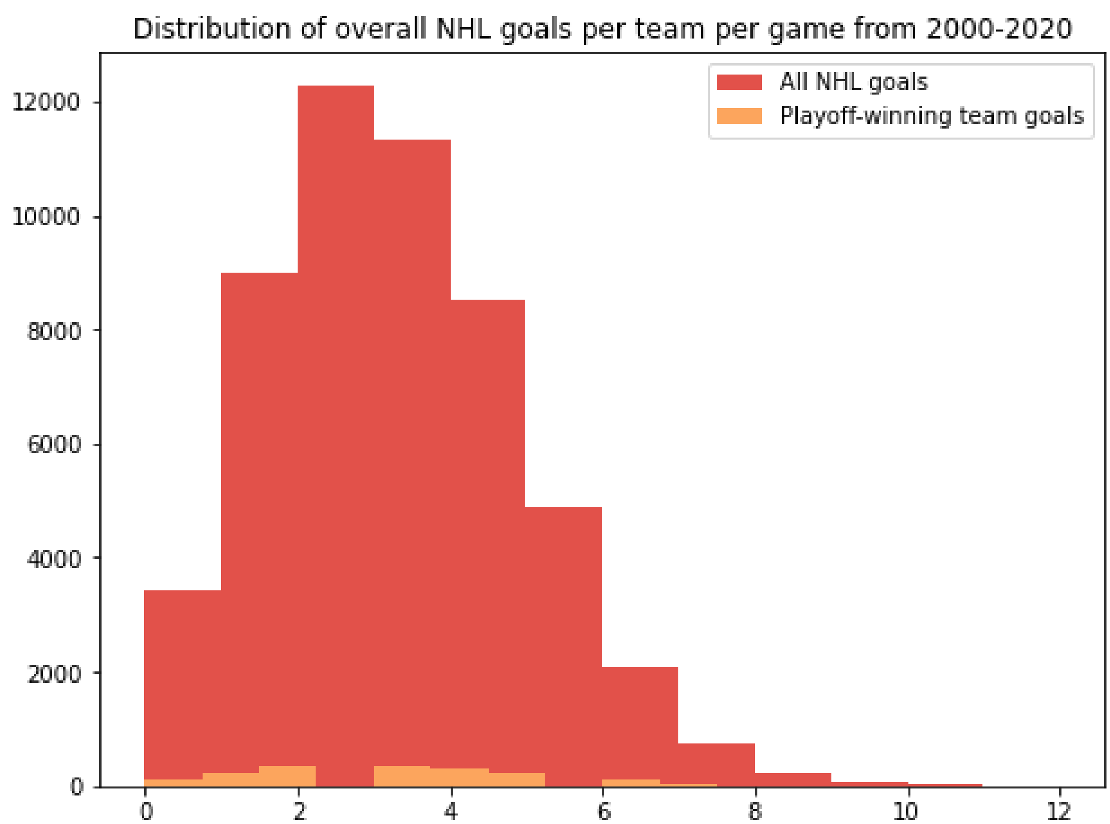
[ ]:
```
      game_id  player_id  team_id  timeOnIce  assists  goals  pim  shots  \
0  2016020045    8473607        4       1504        0      0    0     16
1  2016020045    8473461        4       2011        0      0    0     11
2  2016020045    8470645       16       3598        0      0    0     27
3  2017020812    8468011       24       3696        0      0    0     33
4  2017020812    8475215        7       3539        0      0    0     33

   saves  powerPlaySaves  shortHandedSaves  evenSaves  \
0     12               1                 0         11
1      9               1                 0          8
2     23               2                 0         21
3     30               1                 2         27
4     29               4                 1         24

   shortHandedShotsAgainst  evenShotsAgainst  powerPlayShotsAgainst decision  \
0                        0                13                      3      NaN
1                        0                10                      1        L
2                        0                23                      4        W
3                        3                28                      2        W
4                        1                27                      5        L

   savePercentage  powerPlaySavePercentage  evenStrengthSavePercentage
0       75.000000                33.333333                   84.615385
1       81.818182               100.000000                   80.000000
2       85.185185                50.000000                   91.304348
3       90.909091                50.000000                   96.428571
4       87.878788                80.000000                   88.888889
```

[ ]:
```
winners['team_name'].value_counts()
```

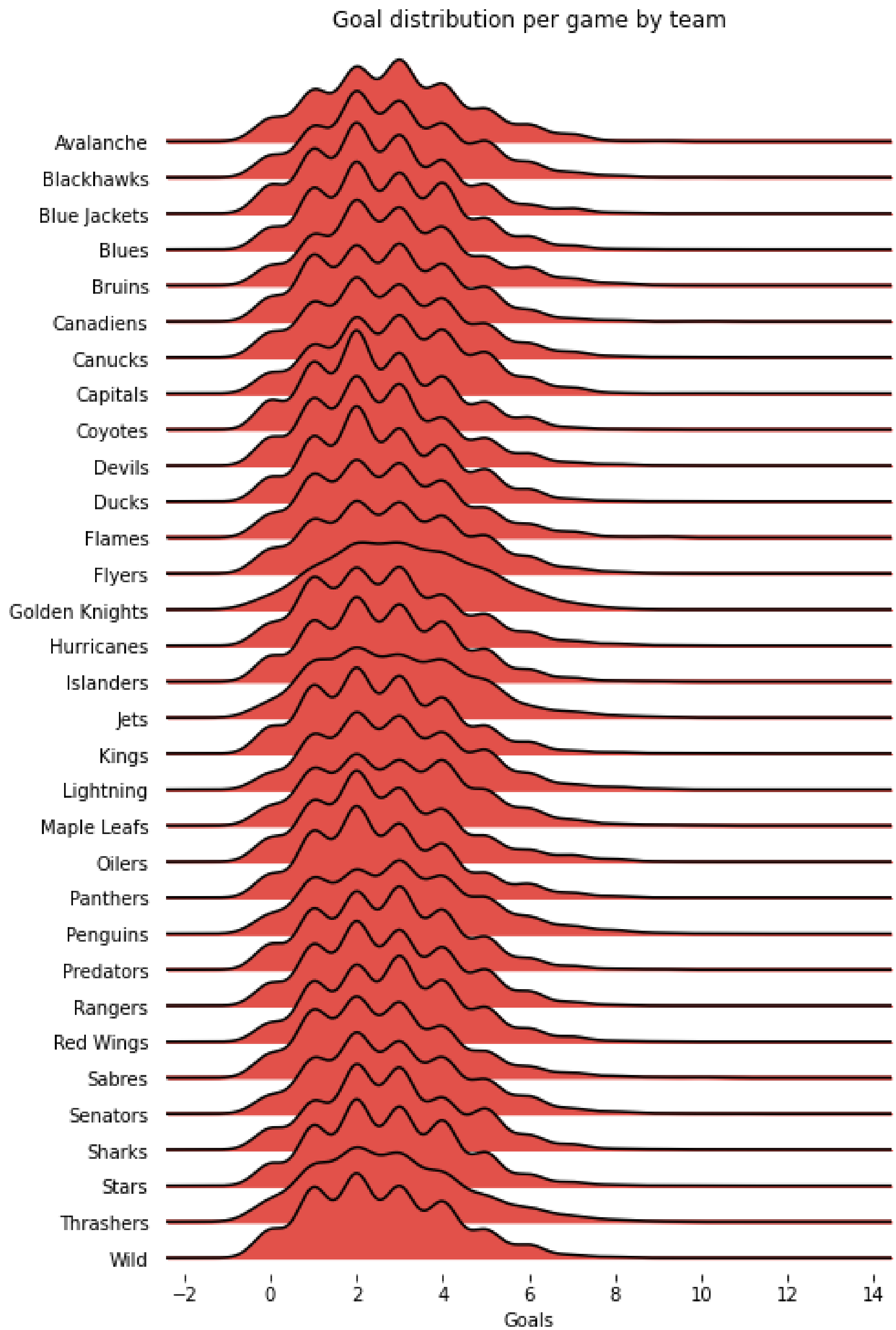
[ ]:
```
Penguins      3
Blackhawks    3
Devils        2
Red Wings     2
Kings         2
Avalanche     1
Lightning     1
None          1
Hurricanes    1
Ducks         1
Bruins        1
```

```
Capitals       1
Blues          1
Name: team_name, dtype: int64
```

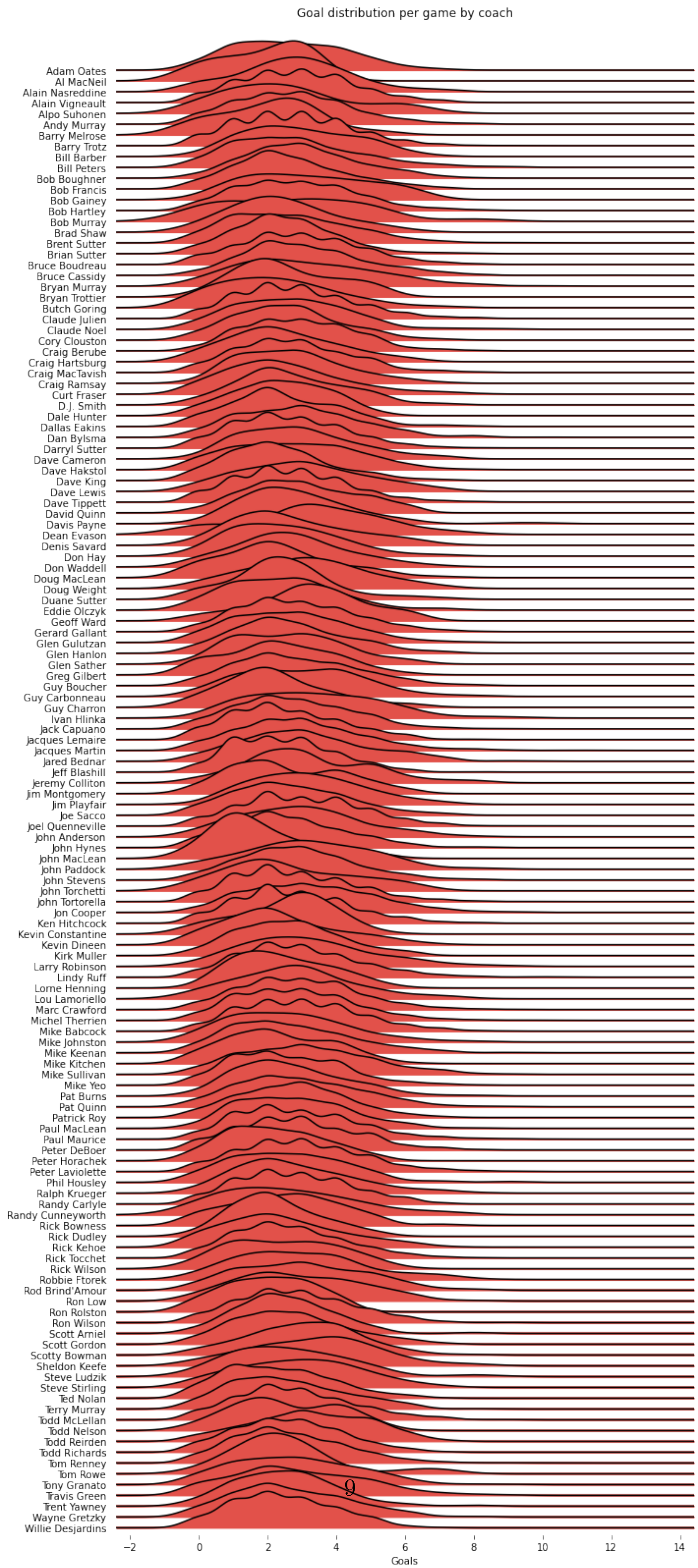
[ ]:
```
plt.figure(figsize=(8, 6))
plt.title('Distribution of overall NHL goals per team per game from 2000-2020')
plt.hist(teams_stats['goals'], bins=12, label='All NHL goals')
plt.hist(teams_stats.query('winseason == 1')['goals'], bins=12,␣
 ↪label='Playoff-winning team goals')
plt.legend()
plt.show()
```



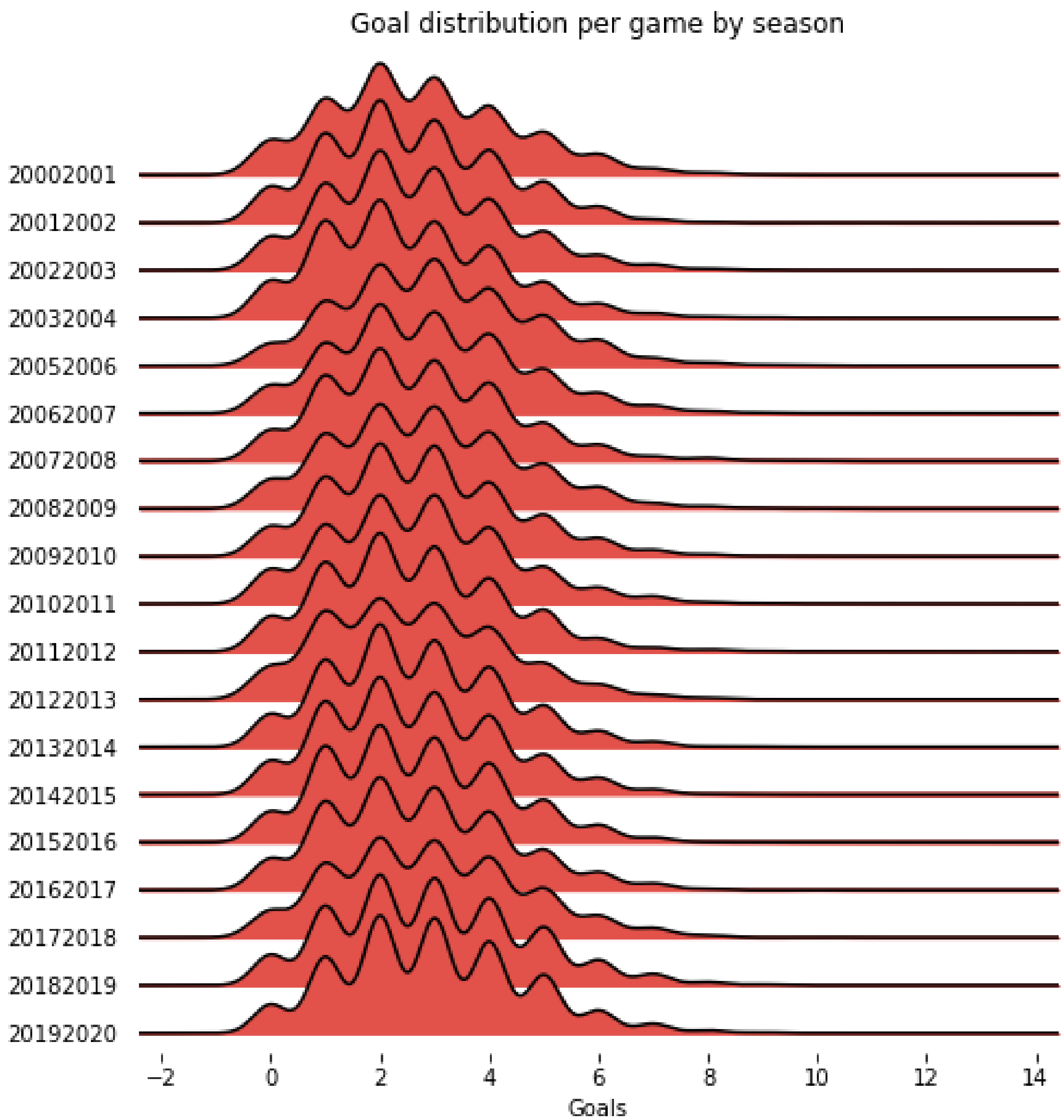
[ ]:
```
joyplot(teams_stats, by='team_name', column='goals', figsize=(7,10));
plt.xlabel('Goals');
plt.title("Goal distribution per game by team");
```


Goal distribution per game by team
Avalanche
Blackhawks
Blue Jackets
Blues
Bruins
Canadiens
Canucks
Capitals
Coyotes
Devils
Ducks
Flames
Flyers
Golden Knights
Hurricanes
Islanders
Jets
Kings
Lightning
Maple Leafs
Oilers
Panthers
Penguins
Predators
Rangers
Red Wings
Sabres
Senators
Sharks
Stars
Thrashers
Wild
−2
0
2
4
6
8
10
12
14
Goals

[ ]:
```
joyplot(teams_stats, by='head_coach', column='goals', figsize=(10,22));
plt.xlabel('Goals');
plt.title("Goal distribution per game by coach");
```

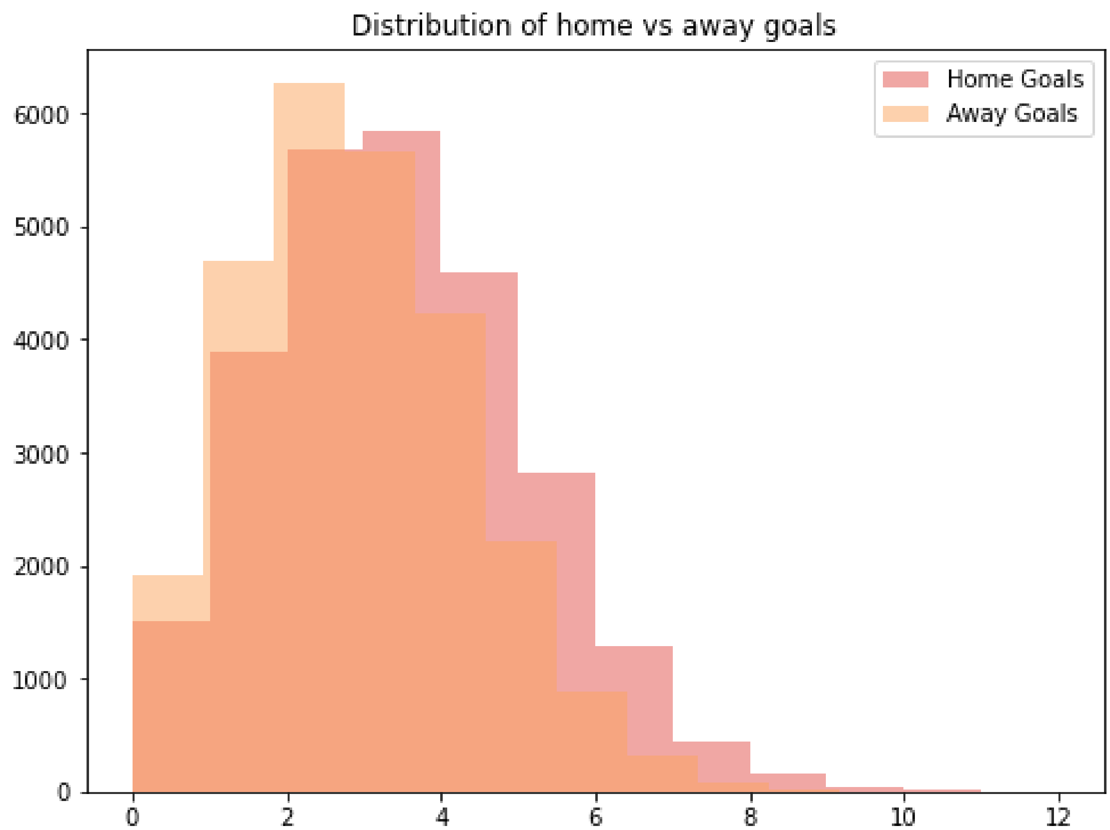

Goal distribution per game by coach
Adam Oates
Al MacNeil
Alain Nasreddine
Alain Vigneault
Alpo Suhonen
Andy Murray
Barry Melrose
Barry Trotz
Bill Barber
Bill Peters
Bob Boughner
Bob Francis
Bob Gainey
Bob Hartley
Bob Murray
Brad Shaw
Brent Sutter
Brian Sutter
Bruce Boudreau
Bruce Cassidy
Bryan Murray
Bryan Trottier
Butch Goring
Claude Julien
Claude Noel
Cory Clouston
Craig Berube
Craig Hartsburg
Craig MacTavish
Craig Ramsay
Curt Fraser
D.J. Smith
Dale Hunter
Dallas Eakins
Dan Bylsma
Darryl Sutter
Dave Cameron
Dave Hakstol
Dave King
Dave Lewis
Dave Tippett
David Quinn
Davis Payne
Dean Evason
Denis Savard
Don Hay
Don Waddell
Doug MacLean
Doug Weight
Duane Sutter
Eddie Olczyk
Geoff Ward
Gerard Gallant
Glen Gulutzan
Glen Hanlon
Glen Sather
Greg Gilbert
Guy Boucher
Guy Carbonneau
Guy Charron
Ivan Hlinka
Jack Capuano
Jacques Lemaire
Jacques Martin
Jared Bednar
Jeff Blashill
Jeremy Colliton
Jim Montgomery
Jim Playfair
Joe Sacco
Joel Quenneville
John Anderson
John Hynes
John MacLean
John Paddock
John Stevens
John Torchetti
John Tortorella
Jon Cooper
Ken Hitchcock
Kevin Constantine
Kevin Dineen
Kirk Muller
Larry Robinson
Lindy Ruff
Lorne Henning
Lou Lamoriello
Marc Crawford
Michel Therrien
Mike Babcock
Mike Johnston
Mike Keenan
Mike Kitchen
Mike Sullivan
Mike Yeo
Pat Burns
Pat Quinn
Patrick Roy
Paul MacLean
Paul Maurice
Peter DeBoer
Peter Horachek
Peter Laviolette
Phil Housley
Ralph Krueger
Randy Carlyle
Randy Cunneyworth
Rick Bowness
Rick Dudley
Rick Kehoe
Rick Tocchet
Rick Wilson
Robbie Ftorek
Rod Brind'Amour
Ron Low
Ron Rolston
Ron Wilson
Scott Arniel
Scott Gordon
Scotty Bowman
Sheldon Keefe
Steve Ludzik
Steve Stirling
Ted Nolan
Terry Murray
Todd McLellan
Todd Nelson
Todd Reirden
Todd Richards
Tom Renney
Tom Rowe
Tony Granato
Travis Green
Trent Yawney
Wayne Gretzky
Willie Desjardins
−2
0
2
4
6
8
10
12
14
Goals

[ ]:
```
joyplot(teams_stats, by='season', column='goals', figsize=(7,7));
plt.xlabel('Goals');
plt.title("Goal distribution per game by season");
```



[ ]:
```
#determine home team advantage

plt.figure(figsize=(8, 6))
plt.title('Distribution of home vs away goals')
```

```
plt.hist(games['home_goals'], bins=12, alpha=.5, label='Home Goals')
plt.hist(games['away_goals'], bins=12, alpha=.5, label='Away Goals')
plt.legend()
plt.show()
```



[ ]:
```
ks_2samp(games['home_goals'], games['away_goals'])
```

[ ]:
```
KstestResult(statistic=0.0677695379349686, pvalue=6.010378102536909e-53,
statistic_location=2, statistic_sign=-1)
```

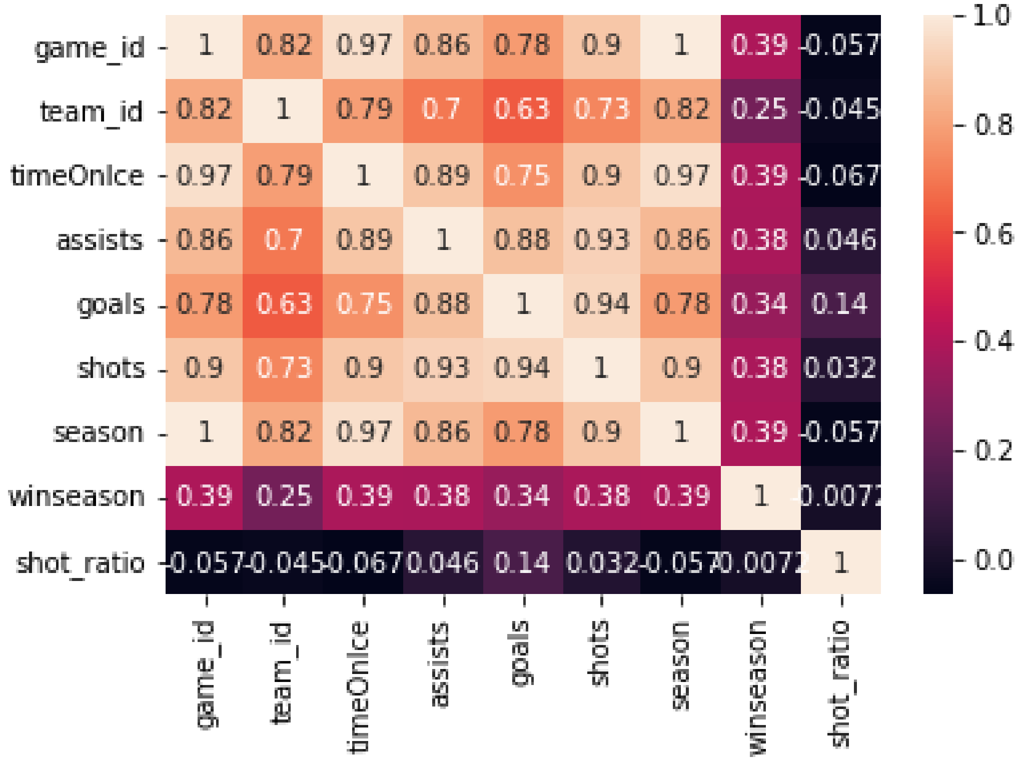
[ ]:
```
scoring_players = skater_stats.groupby('player_id').agg('sum')
scoring_players = scoring_players[scoring_players['goals']>0]
scoring_players['shot_ratio'] = scoring_players['goals']/
 ↪scoring_players['shots']
scoring_players.sort_values('shot_ratio', ascending=False)
```

[ ]:
```
                game_id  team_id  timeOnIce  assists  goals  shots  \
player_id
8479510      4038041826       20       1216        0      2      2
8477836      4026042321       32       1156        0      1      1
```

```
8468036      6018061930     72     1317      0     1     1
8468047      4005042310     24     1001      1     1     1
8473494      6021060384     72      753      0     1     1
…                  …      …        …       …     …     …
8471705    263265707512   2618   111876      9     1    99
8467540    184315887619   2225    86854     24     1   114
8475868    572975993224   9012   225629     36     2   241
8470729    378156888087   2640   181820     20     1   157
8475857    300561076425   3490   127318     15     1   166

                season  winseason  shot_ratio
player_id
8479510       40384040          0    1.000000
8477836       40264028          2    1.000000
8468036       60186021          0    1.000000
8468047       40054007          0    1.000000
8473494       60216024          3    1.000000
…                  …          …         …
8471705     2632893394          0    0.010101
8467540     1843324406          0    0.008772
8475868     5730273254          0    0.008299
8470729     3781908341          0    0.006369
8475857     3005880707          0    0.006024

[2650 rows x 9 columns]
```

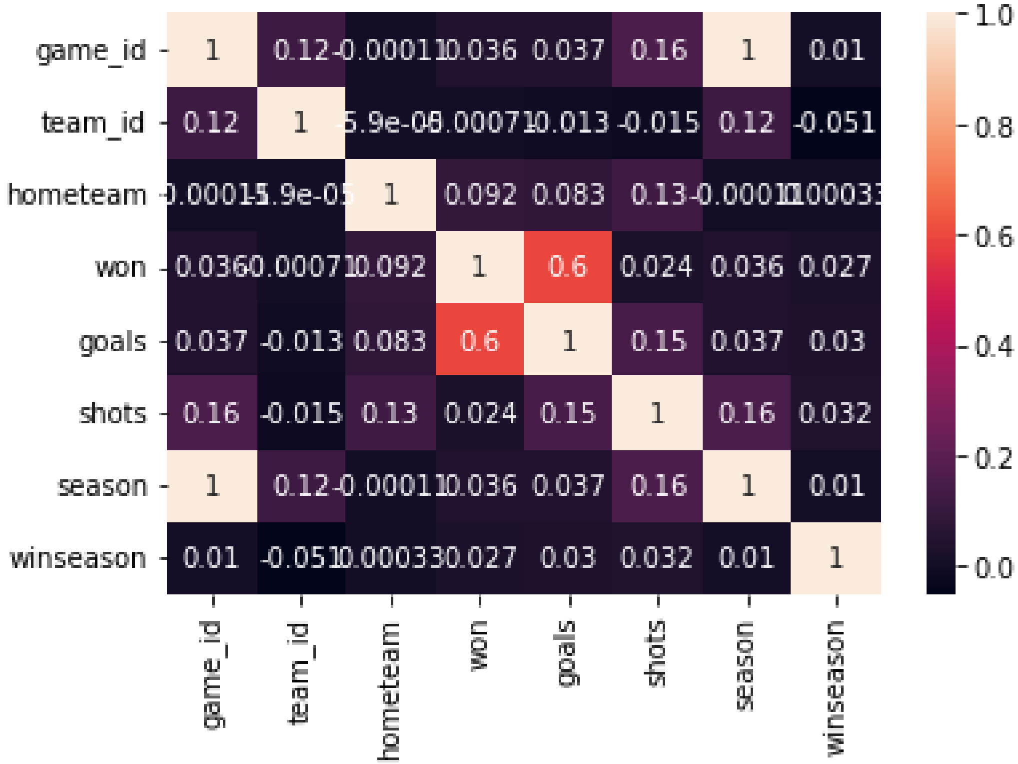
[ ]:
```python
sns.heatmap(scoring_players.corr(), annot=True)
```

[ ]: <AxesSubplot:>



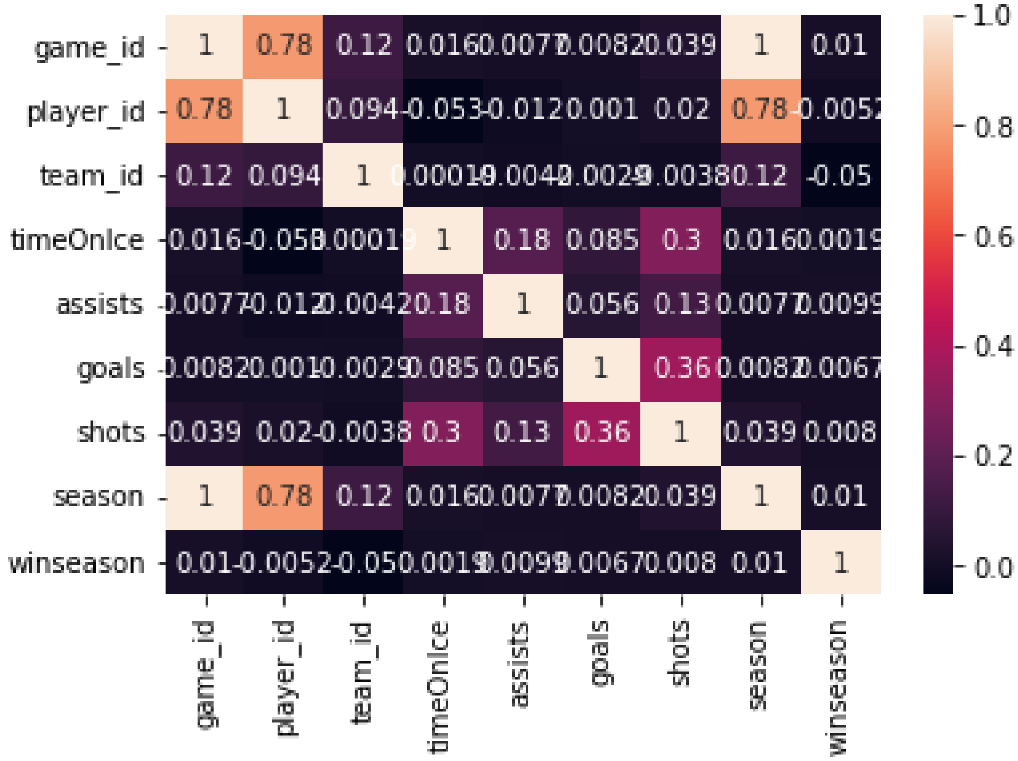
[ ]:
```
sns.heatmap(teams_stats.corr(), annot=True)
```

[ ]: <AxesSubplot:>




```
[ ]: sns.heatmap(skater_stats.corr(), annot=True)
```

```
[ ]: <AxesSubplot:>
```



unsupervised learning teams - label who wins playoff per year, run clustering players - who is on winning teams, run clustering

[ ]:
```python
team_cluster = teams_stats.drop(['settled_in', 'team_name'], axis=1)
dummies = pd.get_dummies(team_cluster['head_coach'])
team_cluster = pd.concat([team_cluster, dummies], axis=1)
team_cluster = team_cluster.drop(['head_coach'], axis=1)
```

[ ]:
```python
team_cluster
```

[ ]:
```
          game_id  team_id  hometeam  won  goals  shots    season  winseason  \
0      2016020045        4         0    0    4.0   27.0  20162017          0
1      2016020045       16         1    1    7.0   28.0  20162017          0
2      2017020812       24         0    1    4.0   34.0  20172018          0
3      2017020812        7         1    0    3.0   33.0  20172018          0
4      2015020314       21         0    1    4.0   29.0  20152016          0
…             …      …         …    …      …      …       …         …
52605  2018030416       19         1    0    1.0   29.0  20182019          0
52606  2018030417       19         0    1    4.0   20.0  20182019          0
52607  2018030417        6         1    0    1.0   33.0  20182019          0
52608  2018030417       19         0    1    4.0   20.0  20182019          0
52609  2018030417        6         1    0    1.0   33.0  20182019          0
```

```
       Adam Oates  Al MacNeil  …  Todd Nelson  Todd Reirden  Todd Richards  \
0               0           0  …            0             0              0
1               0           0  …            0             0              0
2               0           0  …            0             0              0
3               0           0  …            0             0              0
4               0           0  …            0             0              0
…             …         …  …          …            …             …
52605           0           0  …            0             0              0
52606           0           0  …            0             0              0
52607           0           0  …            0             0              0
52608           0           0  …            0             0              0
52609           0           0  …            0             0              0

       Tom Renney  Tom Rowe  Tony Granato  Travis Green  Trent Yawney  \
0               0         0             0             0             0
1               0         0             0             0             0
2               0         0             0             0             0
3               0         0             0             0             0
4               0         0             0             0             0
…             …       …           …           …           …
52605           0         0             0             0             0
52606           0         0             0             0             0
52607           0         0             0             0             0
52608           0         0             0             0             0
52609           0         0             0             0             0

       Wayne Gretzky  Willie Desjardins
0                  0                  0
1                  0                  0
2                  0                  0
3                  0                  0
4                  0                  0
…                …                …
52605              0                  0
52606              0                  0
52607              0                  0
52608              0                  0
52609              0                  0

[52528 rows x 144 columns]
```

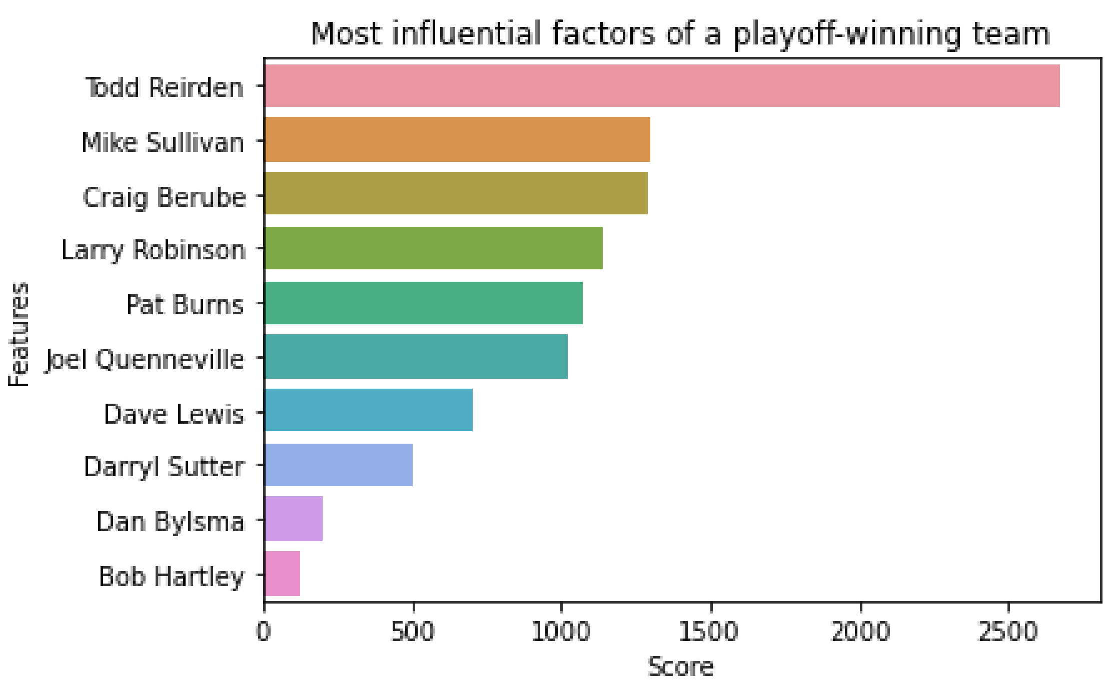
[ ]:
```
from sklearn.feature_selection import SelectKBest
from sklearn.feature_selection import chi2

x = team_cluster.drop(['game_id', 'season', 'team_id', 'winseason'], axis=1)
y = team_cluster['winseason']
```

```
best_features = SelectKBest(score_func=chi2, k=10)
fit = best_features.fit(x,y)

df_scores= pd.DataFrame(fit.scores_)
df_columns= pd.DataFrame(x.columns)

features_scores= pd.concat([df_columns, df_scores], axis=1)
features_scores.columns= ['Features', 'Score']
features_scores = features_scores.sort_values(by = 'Score', ascending=False).
 ↪head(10)
features_scores
```

[ ]:

	Features	Score
131	Todd Reirden	2671.671293
97	Mike Sullivan	1296.999957
30	Craig Berube	1291.769381
87	Larry Robinson	1137.087652
99	Pat Burns	1076.841306
74	Joel Quenneville	1021.887222
43	Dave Lewis	706.152694
39	Darryl Sutter	503.325231
38	Dan Bylsma	202.991597
17	Bob Hartley	126.868730

[ ]:
```
sns.barplot(y='Features', x='Score', data=features_scores).set(title='Most␣
 ↪influential factors of a playoff-winning team')
plt.show()
```

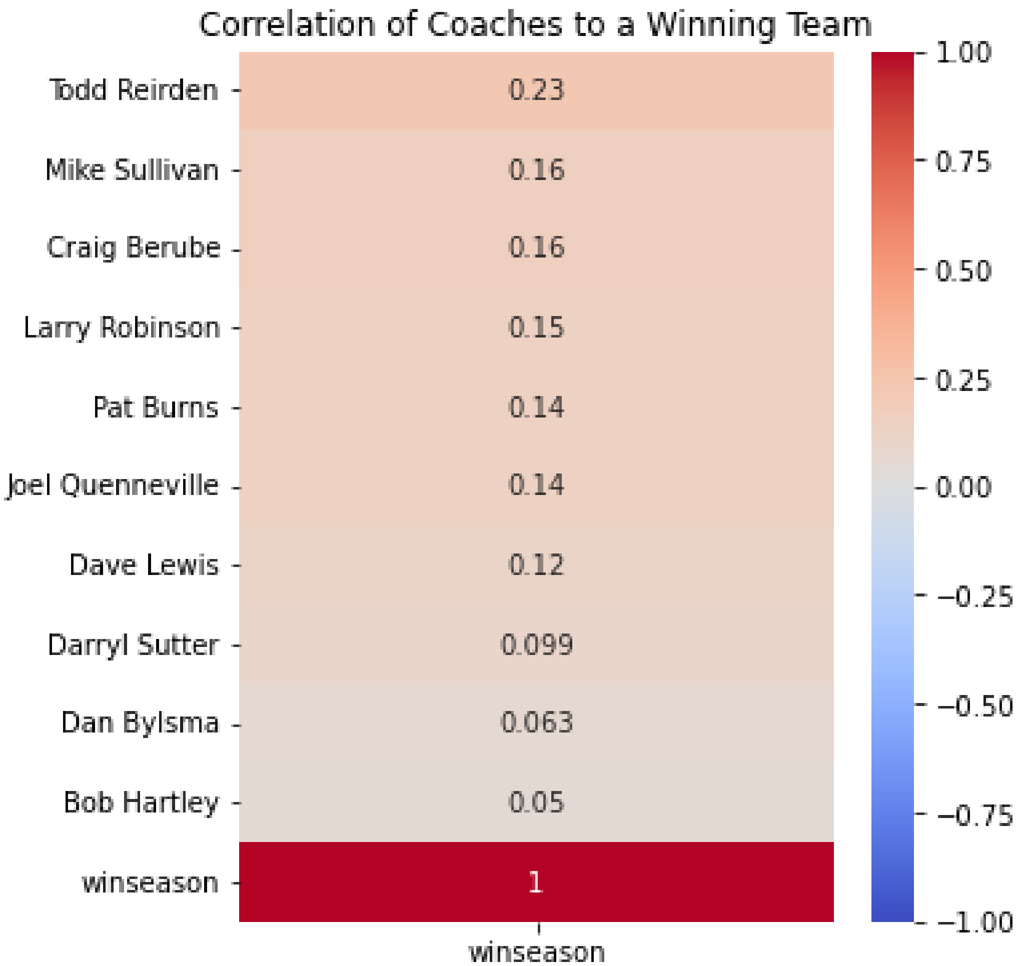


```
[ ]: correlatefeats = features_scores['Features'].tolist()
     correlatefeats.append('winseason')
     correlatefeats
```

```
[ ]: ['Todd Reirden',
      'Mike Sullivan',
      'Craig Berube',
      'Larry Robinson',
      'Pat Burns',
      'Joel Quenneville',
      'Dave Lewis',
      'Darryl Sutter',
      'Dan Bylsma',
      'Bob Hartley',
      'winseason']
```

```
[ ]: plt.figure(figsize=(5,6))
     sns.heatmap(team_cluster[correlatefeats].corr()[['winseason']], annot=True,␣
      ↪vmin=-1, cmap='coolwarm').set(title='Correlation of Coaches to a Winning␣
      ↪Team')
```

```
[ ]: [Text(0.5, 1.0, 'Correlation of Coaches to a Winning Team')]
```



[ ]:
```
team_cluster[team_cluster['Mike Sullivan'] == 1].groupby('season')['winseason'].
    ↪mean()
```

[ ]:
```
season
20032004    0.0
20052006    0.0
20152016    0.0
20162017    1.0
20172018    1.0
20182019    0.0
20192020    0.0
Name: winseason, dtype: float64
```

[ ]:
```
team_cluster[team_cluster['Todd Reirden'] == 1].groupby('season')['winseason'].
    ↪mean()
```

[ ]:
```
season
20182019    1.0
20192020    0.0
Name: winseason, dtype: float64
```

[ ]:
```
team_cluster[team_cluster['Craig Berube'] == 1].groupby('season')['winseason'].
 ↪mean()
```

[ ]:
```
season
20132014    0.0
20142015    0.0
20182019    0.0
20192020    1.0
Name: winseason, dtype: float64
```

[ ]:
```
skater_stats.head()
```

[ ]:
```
      game_id  player_id  team_id  timeOnIce  assists  goals  shots    season  \
0  2016020045    8468513        4        955        1      0      0  20162017
1  2016020045    8476906        4       1396        1      0      4  20162017
2  2016020045    8474668        4        915        0      0      1  20162017
3  2016020045    8473512        4       1367        3      0      0  20162017
4  2016020045    8471762        4        676        0      0      3  20162017

   winseason
0          0
1          0
2          0
3          0
4          0
```

[ ]:
```
skater_stats['winseason'].sum()
```

[ ]:
```
30779
```

[ ]:
```
x = skater_stats.drop(['game_id', 'winseason', 'season', 'team_id',␣
 ↪'player_id'], axis=1)
y = skater_stats['winseason']

best_features = SelectKBest(score_func=chi2, k=3)
fit = best_features.fit(x,y)

df_scores= pd.DataFrame(fit.scores_)
df_columns= pd.DataFrame(x.columns)

features_scores= pd.concat([df_columns, df_scores], axis=1)
features_scores.columns= ['Features', 'Score']
features_scores.sort_values(by = 'Score', ascending=False).head(20)
```

```
[ ]:     Features       Score
     0  timeOnIce  326.765626
     1    assists  100.831432
     3      shots   86.821084
     2      goals   46.076943
```

```
[ ]: correlatefeats = features_scores['Features'].tolist()
     correlatefeats.append('winseason')
     correlatefeats
```

```
[ ]: ['timeOnIce', 'assists', 'goals', 'shots', 'winseason']
```

```
[ ]: win_teams = teams_stats[teams_stats['winseason']==1]
     win_teams.head()
```

```
[ ]:         game_id  team_id  hometeam  won settled_in     head_coach  goals  \
     32   2016021020        5         0    0         OT  Mike Sullivan    3.0
     91   2016020884        5         0    1        REG  Mike Sullivan    3.0
     182  2017020853        5         0    1        REG  Mike Sullivan    4.0
     192  2017020164        5         0    0        REG  Mike Sullivan    1.0
     205  2017020562        5         1    1         OT  Mike Sullivan    4.0

          shots team_name    season  winseason
     32    35.0  Penguins  20162017          1
     91    22.0  Penguins  20162017          1
     182   23.0  Penguins  20172018          1
     192   30.0  Penguins  20172018          1
     205   37.0  Penguins  20172018          1
```

```
[ ]: plt.figure(figsize=(5,6))
     sns.heatmap(skater_stats[correlatefeats].corr()[['winseason']], annot=True,␣
      ↪vmin=-1, cmap='coolwarm').set(title='Correlation of Coaches to a Winning␣
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```
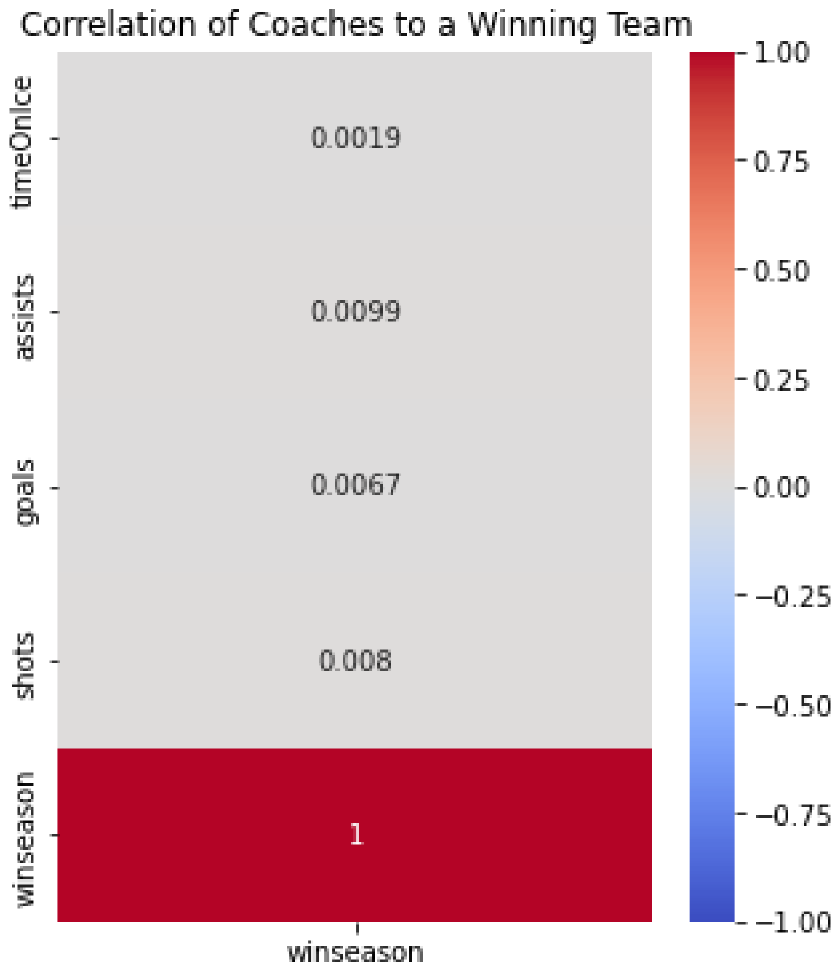

```
[ ]: [Text(0.5, 1.0, 'Correlation of Coaches to a Winning Team')]
```


Correlation of Coaches to a Winning Team
timeOnIce
assists
goals
shots
winseason
0.0019
0.0099
0.0067
0.008
1
winseason
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

```
[ ]:
```